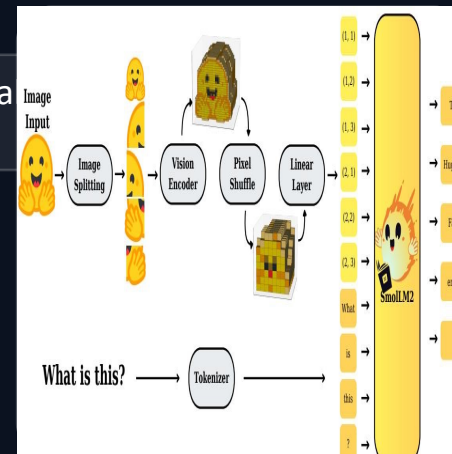


SmolVLM

Redefining Small and Efficient Multimodal Models

Compact VLMs for edge deployment: sub-1GB inference via compute a
+ token compression + long context



Hugging Face & Stanford University

Source: arXiv:2504.05299

Large VLMs Are Powerful but Impractical for Edge Deployment

Core constraint

- State-of-the-art VLMs typically require large memory footprints, blocking mobile/edge use.
- Goal: competitive multimodal capability under tight RAM and compute budgets.
- SmolVLM targets on-device inference and open reproducibility (weights, data, code, app).

0.8 GB

SmolVLM-256M RAM (batch=1)

4.9 GB

SmolVLM-2.2B RAM (batch=1)

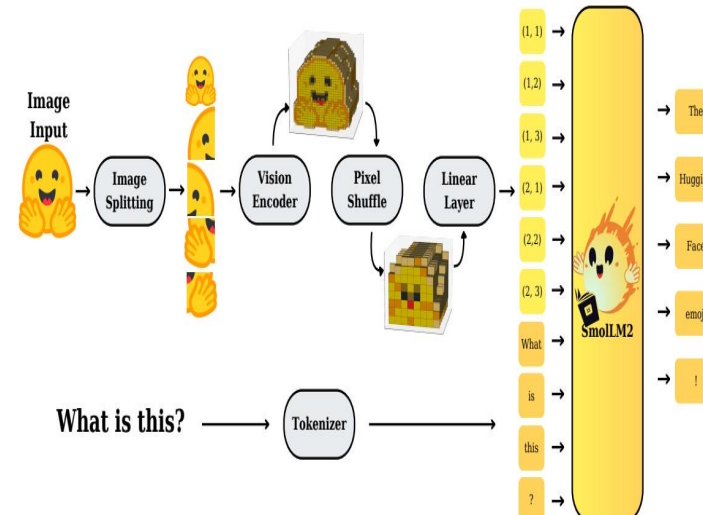
27.7 GB

Molmo-A1B-7B RAM (comparison)

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmoLM2 Backbone

Pipeline (single image)

- Split image into sub-images → encode with SigLIP vision encoder.
- Compress vision features via pixel shuffle (space-to-depth).
- Project to LM embedding space (linear layer).
- Concatenate with text embeddings → SmoLM2 backbone.



Finding 1

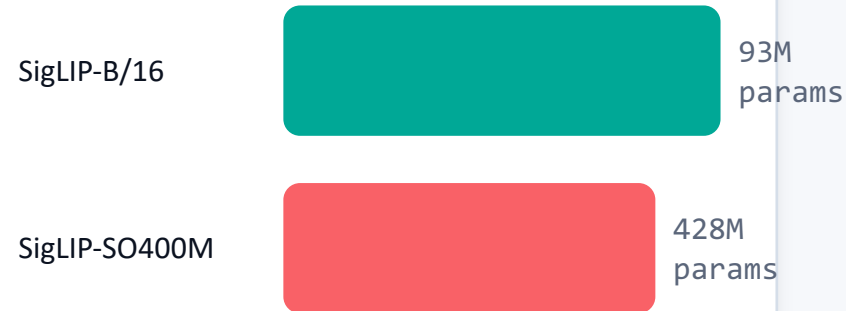
Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Observation

- With a 135M LM, a 93M SigLIP-B/16 outperforms a 428M SigLIP-SO400M.
- Only at ~1.7B LM scale does the larger encoder justify its overhead (~10% params).
- Implication: allocate capacity to the LM earlier; avoid over-spending on vision at small LM scale.

Encoder size vs. compact-LM performance (qualitative)

Per paper: 93M SigLIP-B/16 > 428M SigLIP-SO400M when LM=135M



Note: bars show the reported ordering; the paper does not provide a numeric delta in the brief.

Source: Finding 1 summary in source_brief.md

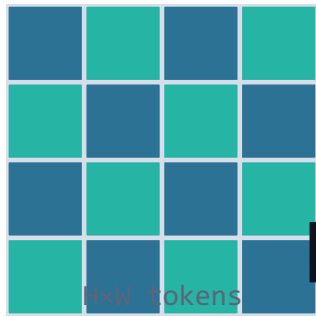
Finding 3

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Pixel shuffle (space-to-depth) at r=4

Token count scales as $r^2 \rightarrow 16\times$ fewer visual tokens

Before



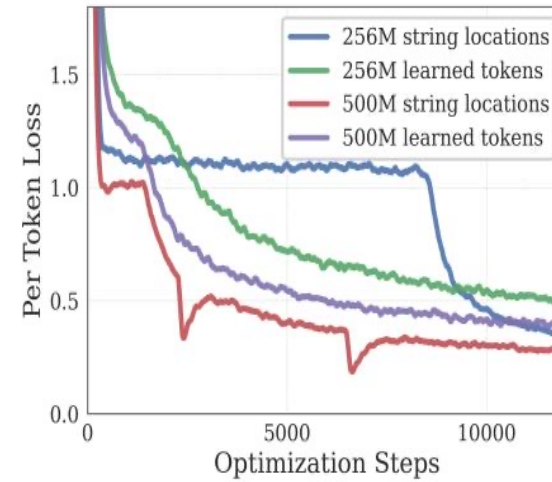
After



16×

shuffle

$H/r \times W/r$
tokens



r=4 helps small VLMs; larger models often prefer r=2.

Source figure: pixel shuffle diagram (arXiv:2504.05299, Fig. 6)

Extending Context from 2K to 16K Tokens Unlocks Higher Image

What changes

- Extend context window from 2K → 16K tokens for significant performance gains.
- Increase RoPE base from 10K → 273K, then fine-tune on mixed long/short-context data.
- 2.2B variant uses 16K context; smaller variants use 8K.

Context scaling recipe

- 1 RoPE base: 10K → 273K
- 2 Fine-tune: mixed long/short
- 3 Deploy: 8K (small) / 16K (2.2B)

2K → 16K context

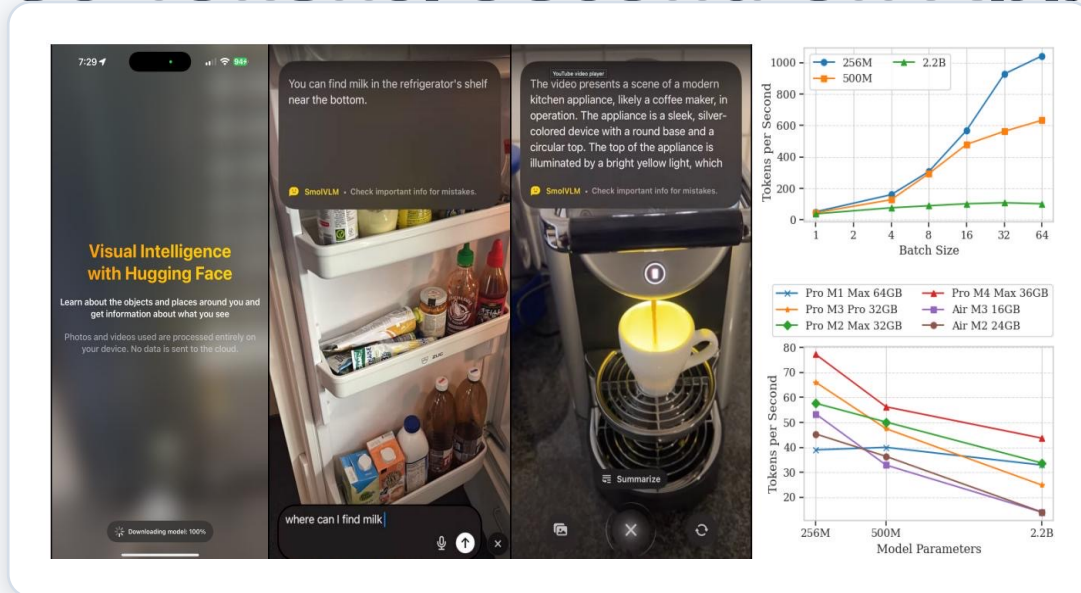
SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Model	Params	Avg (13)	RAM (batch=1)	Best Efficient OS baselines
SmolVLM-256M	256M	44.0%	0.8 GB	Molmo-A1B-7B (27.7 GB RAM)
SmolVLM-500M	500M	51.0%	1.2 GB	InternVL2-2B (Efficient OS for video)
SmolVLM-2.2B	2.2B	59.8%	4.9 GB	Molmo-A1B-7B / InternVL2-2B
Molmo-A1B-7B	7B	(baseline)	27.7 GB	Best Efficient OS (single-image / multi-task)
InternVL2-2B	2B	(baseline)	(not reported)	Best Efficient OS (video benchmarks)

Key finding: SmolVLM-256M outperforms the 300× larger Idefics-80B (per paper).

Averages and RAM from paper summary; Molmo-A1B-7B RAM: 27.7 GB. Benchmarks span single-image, multi-task, and video.

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

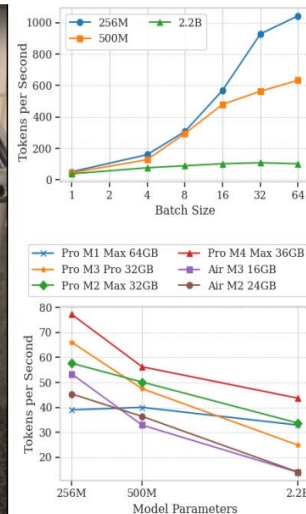


Source figure + throughput bullets in source_brief.md

- Runs fully on-device via HuggingSnap iOS app (no cloud dependency).

Deployment notes

- SmolVLM-256M: up to ~80 tokens/sec on Pro M1 Max 64GB.
- MacBook Air M3 16GB: ~55 tokens/sec (256M).
- At batch=64, 256M reaches nearly 1000 tokens/sec on high-end devices.
- Throughput scales roughly linearly with batch size (smallest model).



Key Takeaways

Design principles for efficient multimodal models

Balance encoder ↔ LM compute

For compact LMs, smaller vision encoders can outperform larger ones.

Use aggressive token compression

Pixel shuffle $r=4$ yields 16× fewer visual tokens; helps small models.

Scale context to unlock resolution

2K → 16K context (RoPE base 10K → 273K) improves performance.

Target sub-1GB inference

SmolVLM-256M runs at 0.8 GB RAM (batch=1) with 44.0% avg score.

Ship open + reproducible

All weights, datasets, code, and HuggingSnap app are open-sourced.

Source: key findings in `source_brief.md`